Scientists Use Ancient Genes to Estimate Contact Between Humans, Neanderthals

Scientists have used ancient genes to learn more details about contact between Neanderthals and humans tens of thousands of years ago.

Results of two recent studies estimate the two groups likely met and began mating about 45,000 years ago.

Modern humans – also known as Homo sapiens – began in Africa hundreds of thousands of years ago and later started spreading to Europe, Asia and other places. Scientists believe that at some point, they met and mated with Neanderthals. The mixing of these groups had a major influence on humans' genetic code.

However, scientists do not know exactly when or how the two groups interacted. But the two new studies provide some additional details about the timing of this contact.

One group of scientists examined genetic material from three female and three male Homo sapiens individuals who lived around 45,000 years ago. Reuters news agency reported that research involved the oldest genes from Homo sapiens ever examined, or sequenced.

Some of the genes came from bones found in a cave in the central German village of Ranis. Other material came from a woman believed to have lived at around the same time in a cave in a mountainous area of what is now the Czech Republic.

This image provided by Max Planck Institute for Evolutionary Anthropology shows an artist's illustration of an ancient human that scientists call Zlatý kůň, whose skull was found in the Czech Republic.(Tom Björklund/Max Planck Institute for Evolutionary Anthropology via AP)

Researchers estimated the period of mixing between Neanderthals and humans at about 49,000 to 45,000 years ago. The findings were recently published in a study in Nature.

A second group of researchers examined genetic material from 300 present-day and ancient Homo sapiens. This included 59 individuals who lived between 2,000 and 45,000 years ago. That study, published in the journal Science, estimated the period of mixing at about 50,500 to 43,500 years ago.

This image provided by National Museum, Prague shows the skull of an ancient human called Zlatý kůň, originally discovered in the Koněprusy caves of the Czech Republic. (Marek Jantač/National Museum, Prague via AP)

The scientists said their new findings on the mixing and mating of the groups suggested the activities happened a little more recently than thought in the past. They believe the contact continued over many generations.

Priya Moorjani was co-writer of the study appearing in Science. She is an assistant professor of molecular and cell biology at the University of California, Berkeley. She told Reuters, “Genetic data from these samples really helps us paint a picture in more and more detail.”

The team noted that it is difficult to know the exact nature of the interactions between Homo sapiens and Neanderthals based on the examined data. The researchers also could not confirm exactly where the mixing and mating happened. However, they believe it was most likely somewhere in the Middle East.

The researchers noted most modern humans still have genetic material from Neanderthals that accounts for an estimated one to two percent of their DNA. They said modern-day genetic traits linked to skin color, hair color and even nose shape can relate back to the Neanderthals. Our genetic makeup also includes links to another group of human ancestors called Denisovans.

Moorjani noted that the history of Neanderthals living outside Africa for thousands of years likely gave them a greater ability to deal with climate and diseases in new environments. “Some of their genes may have been beneficial to modern humans,” she added.

Rick Potts is director of the Smithsonian Institution’s Human Origins program. He was not involved in the new research. Potts told The Associated Press he hopes future genetic studies can help scientists learn even more details about the interactions of Neanderthals and modern humans.

He said, “Out of many really compelling areas of scientific investigation, one of them is: well, who are we?”

I’m Caty Weaver.

Bryan Lynn wrote this story for VOA Learning English, based on reports from The Associated Press, Reuters and Nature.

Quiz - Scientists Use Ancient Genes to Estimate Contact Between Humans, Neanderthals

Vocabulary

genetic code - n. รหัสพันธุกรรม; ข้อมูลจาก DNA หรือ RNA ที่ใช้ในการสร้างโปรตีนของสิ่งมีชีวิต

cave - n. ถ้ำ; โพรงขนาดใหญ่ที่อยู่ด้านข้างของภูเขาหรือใต้ดิน

sample - n. ตัวอย่าง; ปริมาณเล็กน้อยของบางสิ่งบางอย่างที่ให้ข้อมูลเกี่ยวกับสิ่งที่ถูกนำมา

trait - n. ลักษณะเฉพาะ; คุณสมบัติทั้งดีและไม่ดีในลักษณะนิสัยของบุคคล

beneficial - adj. เป็นประโยชน์; ที่เป็นประโยชน์หรือมีประโยชน์

compelling - adj. น่าสนใจอย่างยิ่ง; ที่เรียกร้องความสนใจอย่างมาก

Reading Comprehension Quiz

1. What is the primary purpose of scientists using ancient genes, according to the text?

A) To understand the evolution of modern diseases.

B) To detail the contact between Neanderthals and humans.

C) To identify the oldest human ancestors in Africa.

D) To sequence the complete Neanderthal genome.

2. What did two recent studies estimate regarding the interaction between Neanderthals and humans?

A) They fought over territory around 45,000 years ago.

B) They migrated separately across continents.

C) They likely encountered each other and started interbreeding approximately 45,000 years ago.

D) They shared the same language tens of thousands of years ago.

3. Where are modern humans (Homo sapiens) believed to have originated?

A) Europe

B) Asia

C) The Middle East

D) Africa

4. What significant impact did the mixing of Homo sapiens and Neanderthals have?

A) It led to the extinction of Neanderthals.

B) It accelerated the spread of diseases.

C) It profoundly affected humans' genetic code.

D) It caused modern humans to stop migrating.

5. What specific information about the interaction between Homo sapiens and Neanderthals was previously unknown to scientists?

A) The location of their first meeting.

B) The exact time and manner of their interaction.

C) The physical appearance of their offspring.

D) The reasons for their separation.

6. What made the research by one group of scientists particularly notable?

A) They studied only male Homo sapiens individuals.

B) They discovered entirely new human species.

C) They examined the oldest known Homo sapiens genes ever sequenced.

D) They focused on modern-day genetic material exclusively.

7. Where were some of the ancient genetic materials examined by scientists found?

A) Only in African deserts.

B) In a cave in Ranis, Germany, and the Czech Republic.

C) Exclusively in the Middle East.

D) In South American rainforests.

8. What does the image in Paragraph 8 depict?

A) A photograph of modern human DNA.

B) An artist's illustration of an ancient human named Zlatý kůň.

C) A map showing the migration routes of early humans.

D) A scientific diagram of genetic sequencing.

9. What was the estimated period of mixing between Neanderthals and humans according to the study published in Nature?

A) Approximately 2,000 to 5,000 years ago.

B) Around 49,000 to 45,000 years ago.

C) Between 50,500 and 43,500 years ago.

D) Hundreds of thousands of years ago.

10. How many individuals, both present-day and ancient Homo sapiens, were examined by the second group of researchers?

A) 59

B) 45,000

C) 300

D) 2,000

11. What was a key implication of the scientists' new findings regarding the mixing and mating of Neanderthals and humans?

A) The activities ceased abruptly after a short period.

B) The events occurred much earlier than previously believed.

C) The contact happened more recently than previously thought and continued for many generations.

D) The mixing only involved a single generation.

12. According to Priya Moorjani, what is the benefit of using genetic data from these samples?

A) It provides a definitive timeline for Neanderthal extinction.

B) It allows scientists to create a more detailed understanding of the past.

C) It helps to identify future human evolutionary paths.

D) It proves that Neanderthals had superior intelligence.

13. What did the researchers suggest was the most probable location for the mixing and mating between Homo sapiens and Neanderthals?

A) Africa

B) Europe

C) The Middle East

D) Asia

14. What percentage of their DNA are most modern humans estimated to have from Neanderthals?

A) Less than one percent.

B) Exactly two percent.

C) One to two percent.

D) More than five percent.

15. What reason did Moorjani suggest for Neanderthal genes potentially being beneficial to modern humans?

A) Neanderthals had a stronger social structure.

B) Their long history outside Africa likely improved their adaptation to new climates and diseases.

C) They possessed superior hunting techniques.

D) Their genes boosted intelligence in modern humans.